

3-Level Geometric Modeler: CSG, Subdivision Trees and Boundary Representation

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I. INTRODUCTION

Solid modeling is distinguished from related areas of geometric modeling and computer graphics by its emphasis on physical fidelity. Together, the principles of geometric and solid modeling form the foundation of computer-aided design and in general support the creation, exchange, visualization, animation, interrogation, and annotation of digital models of physical objects.

We will describe a polyhedron as three sets : (V, E, F) . V represents set of vertices of the polyhedron. E represents the set of edges of the polyhedron. F represents the set of faces of the polyhedron.

II. DEFINITIONS

2.1 Regularized set

A set $S \subset \mathbb{R}^3$ is a regularized set iff $S = \overline{\text{int}(S)}$. All of our operations are on Regularized sets.

2.2 Regularized set operation on regularized sets A and B

- $A \cup^* B = \overline{\text{int}(A \cup B)}$
- $A \cap^* B = \overline{\text{int}(A \cap B)}$
- $A -^* B = \overline{\text{int}(A - B)}$
- $A^c * B = \overline{\text{int}(A^c)}$

2.3 Solid

A solid S is defined as a regularized subset of $\mathbb{R}^3$ ($S \subset \mathbb{R}^3$)

III. CSG OBJECTS

Just as a polyhedron can be described as the intersection of half spaces, a CSG (Constructive Solid Geometry) model describes a polyhedron as Boolean operation on primitives which are the simplest solid objects used for the representation. The Boolean operations on primitives are intersection, union, difference and complement.

3.1 Primitives

Primitives in CSG models are all semialgebraic sets as follows:

1. Half Space: $H = \{(x, y, z) | a * x + b * y + c * z - d \geq 0\}$
2. Sphere : $S = \{(x, y, z) | (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 - r^2 \leq 0\}$.

IV. SUBDIVISION TREES

1. Box: B is a box iff $B \subset \mathbb{R}^3$, $B = \{(x, y, z) | x_1 \leq x \leq x_2, y_1 \leq y \leq y_2, z_1 \leq z \leq z_2\}$, which is a regularized subset of $\mathbb{R}^3$
2. Elementary Box of a polyhedron P :
 - (a) Empty Box: a box B is an Empty Box iff $B \cap P = \emptyset$.
 - (b) Full Box: a box B is a Full Box of Polyhedron P iff $B \cap P = B$
 - (c) Vertex Box: a box B is a Vertex Box iff there exists only one vertex $v \in V \cap B$, s.t. $\forall e \in E$, if $e \cap B \neq \emptyset$, v bounds e , and $\forall f \in F$, if $f \cap B \neq \emptyset$, v bounds F .

- (d) Edge Box: a box B is an Edge Box iff $V \cap B = \emptyset$ and there exists only one edge $e \in E$, $e \cap B \neq \emptyset$, $\forall f \in F$, if $f \cap B \neq \emptyset$, e bounds f .
 - (e) Face Box: a box B is a Face Box iff $\forall e \in E$, $e \cap B = \emptyset$, and there exists one face $f \in F$, and $f \cap B \neq \emptyset$.
3. Subdivision tree : A subdivision tree in $\mathbb{R}^3$ is an octree, in which each node points to a box and we divide the each box pointed by a inner node into 8 children pointing to 8 subboxes.
4. Adequate subdivision tree of a polyhedron P : A subdivision tree whose leaves are all elementary boxes of the polyhedron P .
5. Subdivision Rules of a box B_0 corresponding to a polyhedron P :
- (a)
 - (b)